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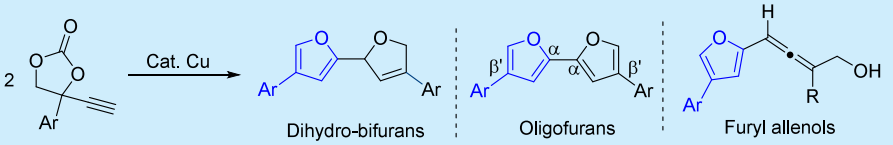
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De Novo Synthesis of Oligofurans via Copper-Catalyzed Dimerization of Ethynylethylene Carbonates

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- Propargyl alcohols act as both an electrophile and a nucleophile (1st time)
- Controlled synthesis of dihydro-bifurans, oligofurans and allenols
- The rapid approach to β',β'-diaryl-α,α-oligofuran monomer (1st time)

ABSTRACT: Disclosed here is an efficient Cu-catalyzed de novo synthesis of oligofurans, a type of valuable core unit in organic materials. From the readily available ethynylethylene carbonates, the dimerization process allowed convenient access to differently substituted oligofurans that could not have been easily synthesized before. In contrast to the conventional strategies, this protocol does not rely on preinstallation of cross-coupling functionalities or presynthesis of advanced enyne precursors. Brief studies of their photophysical properties also indicated the great potential of our protocol in tuning these properties for applications. Mechanistic studies not only provided important insights into the reaction pathway but also allowed the identification of another protocol for the synthesis of furan-substituted allenylic alcohols.

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Oligofurans and oligothiophenes are important core units in organic dyes and related semiconductor molecules (Figure 1a).[1−10] The structurally rigid electron-rich π-conjugation system has a critical influence on the photophysical and electrochemical properties of the resulting materials. Specifically, oligofuran-based organic materials not only have superior electron transport performance but also possess good biodegradability, making them highly attractive in the development of environmentally friendly electronic devices for large-scale applications.[11−13]

Over the past few decades, the synthesis of oligofurans and related structures has primarily relied on tedious procedures involving repetitive cross-coupling from the corresponding furyl electrophiles and nucleophiles. The less friendly stannyl furan species were often used as nucleophilic partners, which also posed significant concerns to the environment and human health (Figure 1b).[1−7,14,15] The cross-coupling products were sometimes contaminated by homocoupling products with similar polarity, thus leading to challenges in purification. Moreover, due to the strong nucleophilicity of the α-positions in furan rings, the conventional synthesis of oligofurans has been limited to α',α'-diaryl-α,α-oligofurans **II**. To overcome the above drawbacks, tremendous progress has been achieved in developing more sustainable and efficient strategies.[16−22] Among them, Zhu and co-workers have recently reported an elegant de novo strategy featuring the cyclization of a linear enyne-conjugated aldehyde precursor in the presence of a Brønsted acid or π-Lewis acid catalyst.[21,22] While such protocols are highly atom-economic and environmentally benign, the synthesis of those advanced linear precursors may take a few steps (Figure 1c).

Dimerizations of terminal alkynes are important tools in organic synthetic, particularly the "head-to-head" or "head-to-tail" dimerizations, which serve as key strategies for synthesizing 1,3-enynes.[23−26] However, due to the propargylic leaving group, the metal-catalyzed cleavage of the propargylic leaving group often precedes the deprotonation at the terminal position, thereby forcing the propargyl substitution to be the major pathway for terminal propargylic electrophiles (Figure 1d).[27−32] As a result, the development of dimerization of propargyl electrophiles involving cleavage of the leaving group has remained highly challenging, thus impeding advances in this realm.

Aiming to further simplify the assembly line, we envisioned the possibility of intermolecular dimerization of readily

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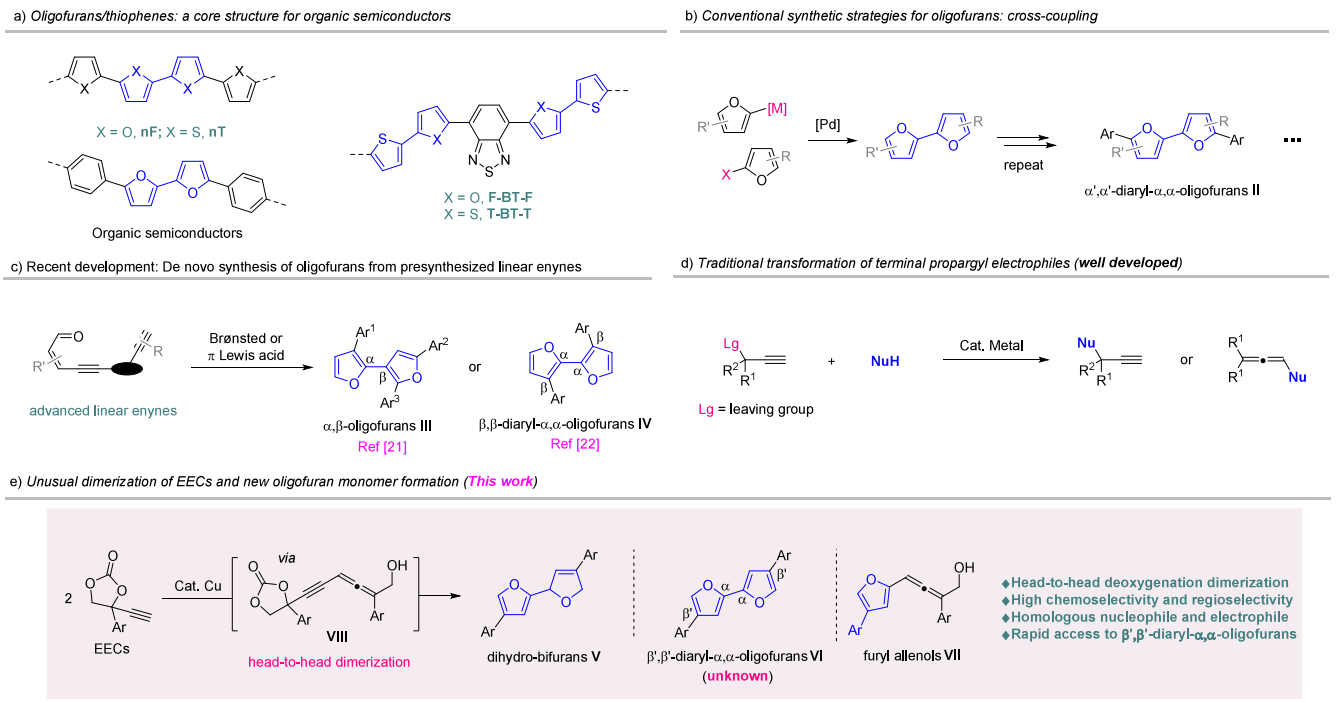


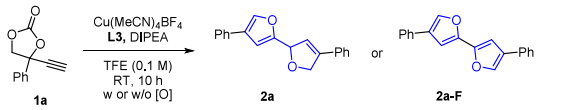
Figure 1. Introduction to the synthesis of oligofurans and reaction design.

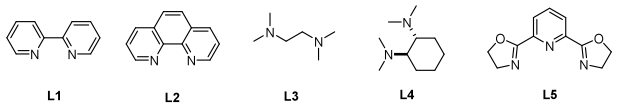
available simple alkyne substrates, like ethynylethylene carbonates (EECs),[33−37] which features the formation of both furan rings and the C−C bond between them in the same process (Figure 1e). To the best of our knowledge, this represents the first example of EECs simultaneously acting as both nucleophiles and electrophiles at the same reaction site, affording furyl dihydrofurans **V** via head-to-head deoxygenative dimerization of EECs. These products can be transformed into oligofurans **VI** in one pot, which holds immense promise in the development of functional materials. Additionally, we demonstrated a solvent-controlled dimerization approach to construct diverse furyl allenols **VII** via an interrupted pathway.

With **1a** as the model substrate and a Cu(I) salt as the catalyst, we were delighted to determine that dimerization product **2a** could be directly generated (Table 1). Systematic evaluation of different parameters indicated that, with $Cu(MeCN)_4BF_4$ as the catalyst, diamine **L3** as the ligand, diisopropyletheylamine (DIPEA) as the base, and trifluoroethanol (TFE) as the solvent, the desired product could be generated in 82% yield at room temperature (entry 1 and more details in the Supporting Information). Other ligands, such as bipyridine **L1**, phenanthroline **L2**, cyclohexanediamine **L4**, and PyBox **L5**, provided decreased yields (entries 2−5, respectively). Other copper salts, including Cu(OTf) and $Cu(OTf)_2$, proved to be much less effective (entries 6 and 7, respectively). Interestingly, less polar solvents (THF, DCM, and toluene) were completely unsuitable for this reaction, resulting in no reaction (entry 8). KOH and DBU were less effective bases, but Et_3N led to a slightly higher yield (entries 9 and 10). Increasing the temperature to 60 °C could further improve the efficiency (91% yield, entry 11).

We were also curious about the possibility of the direct formation of bis(furan) from the same pot in the presence of an oxidant. Thus, various oxidation conditions were examined on the crude product (see the Supporting Information for more details). To our delight, a one-pot reaction employing elemental sulfur in DMF efficiently furnished oligofuran **2a-F** in 57% isolated yield, thus representing a convenient de novo strategy for the synthesis of oligofurans (entry 12).

Table 1. Evaluation of the Reaction Conditions for Alkyne Dimerization[a]





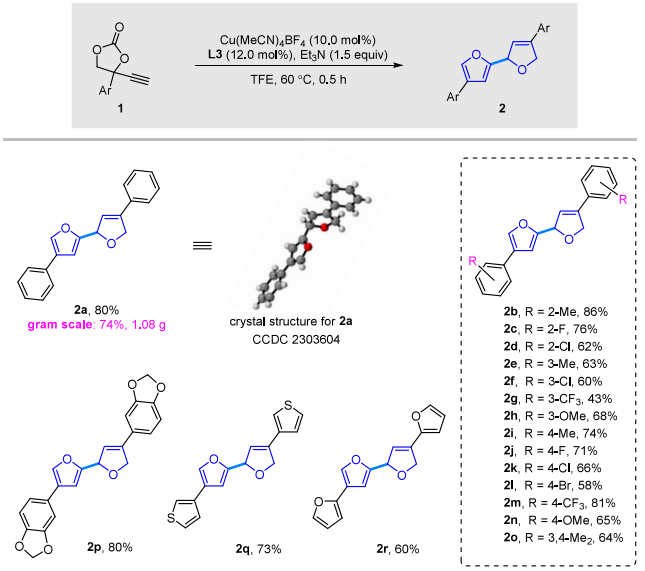
entry	variation of conditions	yield (%)	product
1	none	82	**2a**
2	**L1** as the ligand	30	**2a**
3	**L2** as the ligand	39	**2a**
4	**L4** as the ligand	66	**2a**
5	**L5** as the ligand	31	**2a**
6	$Cu(OTf)_2$ as the copper salt	15	**2a**
7	Cu(OTf) as the copper salt	20	**2a**
8	DCM, toluene, or THF as the solvent	NR	**2a**
9	KOH or DBU as the base	45−47	**2a**
10	Et_3N as the base	84	**2a**
11[b]	Et_3N as the base, 60 °C	91	**2a**
12[c]	with an oxidant	57	**2a-F**

[a]Reaction conditions: **1a** (0.1 mmol), $Cu(MeCN)_4BF_4$ (10.0 mol %), a ligand (12.0 mol %), a base (1.5 equiv), TFE (1.0 mL), RT, 10 h. The yield was determined by ^1H NMR analysis of the crude product with CH_2Br_2 as an internal standard. [b]Run for 0.5 h. [c]One-pot oxidation conditions: **1a** (0.2 mmol), $Cu(MeCN)_4BF_4$ (10.0 mol %), **L3** (12.0 mol %), Et_3N (1.5 equiv), TFE (2.0 mL), 60 °C, 0.5 h; then evaporation followed by addition of sulfur powder (0.5 equiv) and DMF (1.0 mL), 160 °C, 3 h. Isolated yield.

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Under the above optimized conditions, a range of EECs were examined for this Cu-catalyzed dimerization process (Scheme 1). Substitution at the *ortho*, *meta*, and *para* positions

Scheme 1. Scope of the Dimerization of EECs[a]



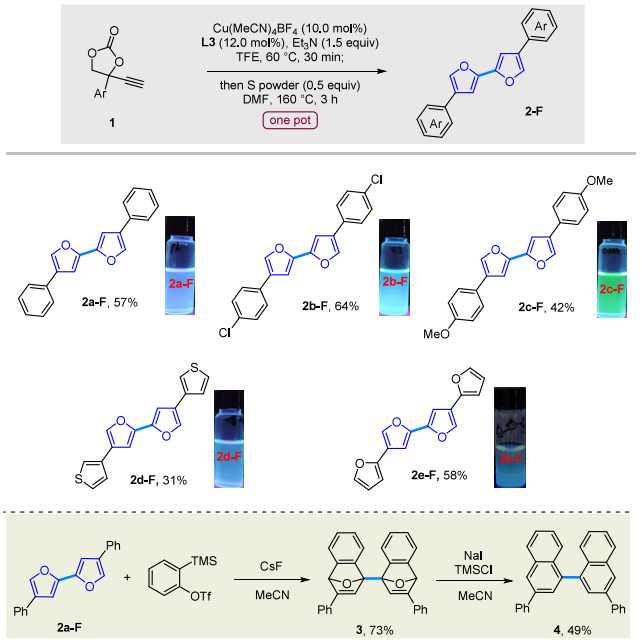
[a]Standard conditions: **1** (0.2 mmol), $Cu(MeCN)_4BF_4$ (10.0 mol %), **L3** (12.0 mol %), Et_3N (1.5 equiv), TFE (2.0 mL), 60 °C, 0.5 h. Isolated yields.

of the side benzene rings successfully delivered the desired products in good to high yields with excellent selectivity. Substrates with different electronic properties were suitable, with a limited influence on the reaction efficiency. The structure of **2a** was confirmed by X-ray crystallography (CCDC 2303604). It is worth noting that heteroarenes, such as thiophene and furan, could also be incorporated into the substrates, which resulted in the formation of oligofuran/thiophenes **2q** and **2r** bearing four such rings connected consecutively. Compared with the conventional strategies that couple each ring in a stepwise manner, our protocol is much more convergent and efficient.

We also examined the generality of the one-pot oxidative protocol for the direct formation of oligofurans (Scheme 2). A series of oligofurans (**2a-F**–**2e-F**) were successfully obtained from the corresponding EECs. It is worth noting that this is the first synthesis of **2d-F** and **2e-F**, where four consecutive furan/thiophene rings are connected at positions 2 and 3. It would not be straightforward to access such skeletons by conventional strategies.[18−20] Notably, these highly conjugated structures themselves show different colors, suggesting their potential in photophysical applications. To further demonstrate the utility of the oligofuran products, we also studied a Diels–Alder reaction between **2a-F** and *in situ*-generated benzyne. This reaction initially afforded adduct **3**, but upon treatment of TMSCl and NaI, it was further transformed into substituted binaphthyl **4**, a valuable unit in organic synthesis and materials science.[38−41]

The capability of our protocol in conveniently accessing differently substituted oligofurans promoted us to quantitatively study their photophysical properties. As shown in Figure 2a, the UV–vis absorption properties of these structures were examined. While their absorption wavelengths turned out to be

Scheme 2. One-Pot Oxidative Synthesis of Oligofurans and Further Derivatization[a]



[a]Standard conditions: **1** (0.2 mmol), $Cu(MeCN)_4BF_4$ (10.0 mol %), **L3** (12.0 mol %), Et_3N (1.5 equiv), TFE (2.0 mL), 60 °C, 0.5 h; then evaporation followed by addition of sulfur powder (0.5 equiv) and DMF (1.0 mL), 160 °C, 3 h. Isolated yield.

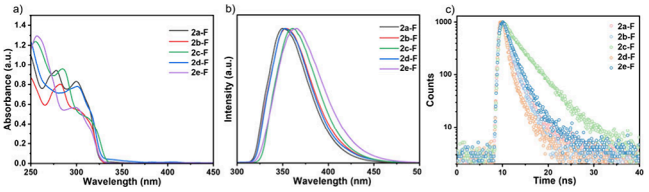


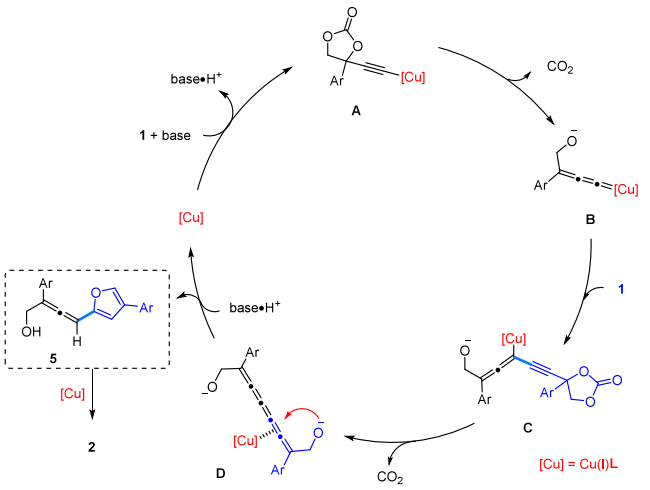
Figure 2. Photophysical properties of the oligofuran products. (a) UV–vis absorption spectra in DCM (1×10^{-5} M). (b) Emission spectra in DCM (1×10^{-5} M). (c) PL lifetime in DCM (1×10^{-5} M).

similar in the range of 250–320 nm, electron-rich **2e-F** has a notably stronger absorption compared to other oligofurans. The emission properties of these compounds were also investigated. A significant red-shift for **2e-F** and **2c-F** was observed, but thiophene-substituted oligofuran **2d-F** did not exhibit a pronounced effect on the optical properties (Figure 2b). Photoluminescence (PL) lifetimes of these oligofurans were also determined (Figure 2c). The PL lifetimes of **2a-F**–**2e-F** are 1.02, 1.09, 4.66, 1.00, and 1.77 ns, respectively. The solubility of oligofuran **2e-F** was found to be higher than those of other oligofurans/thiophenes, indicative of better processability.[18−20] These preliminary results indicated that our products hold promise for the development of new organic materials. The convenient synthesis from simple substrates is expected to allow further modification and optimization of their properties.

A possible mechanism is depicted in Scheme 3. In the presence of a base, the substrate alkyne can easily generate copper(I) acetylide species **A**, which undergoes decarboxylation to form copper allenylidene **B** by eliminating CO_2. Subsequently, another molecule of substrate **1** may serve as a

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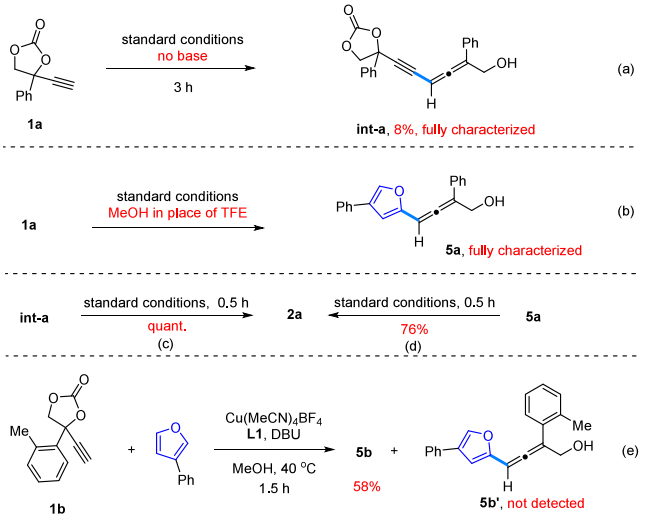
Scheme 3. Proposed Mechanism



nucleophile to attack the carbene motif, generating conjugated allenyl alkyne **C**. A similar decarboxylation may take place to form cumulative allene **D**, in which copper(I) serves as a π-acid to trigger a cyclization from one alkoxide motif together with isomerization and protodecupration to form furan-substituted allenylic alcohols **5**. Then, the Cu(I) catalyst can further activate the allene motif to trigger another intramolecular cyclization to form observed dimerization product **2**.[42−44]

To gain more insight into the proposed mechanism, we carried out a series of control experiments (Scheme 4). In the

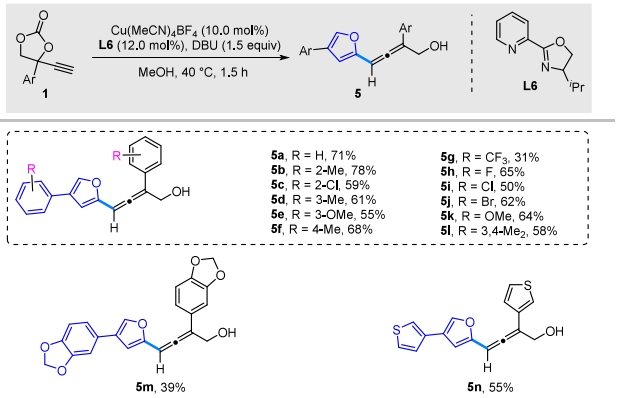
Scheme 4. Control Experiments



absence of a base, we were able to isolate a small amount of allenyl alkyne **int-a** (Scheme 4, a), presumably from direct protonation of intermediate **C**. Moreover, with methanol as a solvent, the reaction of **1a** could generate **5a** as the major product (Scheme 4, b). Importantly, both **int-a** and **5a** are chemically competent for the generation of dimerization product **2a** under the standard conditions. **int-a** was cleanly converted into observed dimerization product **2a** under the standard conditions (Scheme 4, c). Similarly, the reaction of **5a** provided **2a** in 76% yield (Scheme 4, d). We were also curious about whether the cyclization takes place prior to the linking of two molecules, which would involve coupling with the initially formed furan with another molecule of alkyne. Thus, 3-phenylfuran and substrate **2b** was mixed under the modified conditions that are specific for the generation of the direct coupling intermediate, furyl allenylic alcohol. Unfortunately, only homodimerization product **5b** was observed, but no crossover product **5b′**. This observation is consistent with the proposed mechanism involving cyclization after intermolecular C−C bond formation (Scheme 4, e).

In fact, furan-substituted allenylic alcohols could be useful synthetic building blocks, but general access to furan-substituted allenylic alcohols has previously been lacking.[45−49] The study presented above suggested that these compounds could be generated from this dimerization reaction. Thus, we optimized the conditions specifically for the formation of this type of product on purpose (see the Supporting Information for details). Indeed, with **L6** as the ligand and MeOH as the solvent, the reaction of **1a** could afford allenol **5a** in 71% isolated yield (Scheme 5). The scope of this protocol was also briefly examined. Different substituted EECs also proceeded smoothly to form the desired furyl allenol products as the major products.

Scheme 5. Synthesis of Furyl Allenols 5[a]



[a]Standard conditions: **1** (0.2 mmol), $Cu(MeCN)_4BF_4$ (10 mol %), **L6** (12 mol %), DBU (1.5 equiv), MeOH (2.0 mL), 40 °C, 1.5 h. Isolated yield.

In summary, we have developed a new de novo synthesis of oligofurans from readily available alkyne substrates. In contrast to the conventional strategies, no preinstallation of cross-coupling functionalities (such as halide and stannyl groups) or presynthesis of the advanced linear enyne precursors is needed. The intrinsically controlled C−C bond formation and cyclization allowed convenient access to oligofurans, a type of valuable subunit in organic materials. The photophysical properties of these products were studied, suggesting the potential of this protocol in generating previously limitedly accessible but useful conjugated molecules of this type. Control experiments provided important insight into the reaction mechanism, which likely involves the formation of intermolecular C−C bonds prior to cyclization. The related study also permitted the establishment of a protocol for the synthesis of furan-substituted allenylic alcohols, another family of important building blocks.

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■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c02955.

Experimental procedure, characterization data, crystal data for **2a**, and NMR spectra (PDF)

Accession Codes

Deposition Number 2303604 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Author Contributions

Z.H. and B.Z. contributed equally to this work. Z.H. and B.Z. designed the experiments and analyzed the data. L.W. and L.X. contributed to the preparation of substrates and ligands. Y.L., X.Z., and J.Y. performed the photophysical test. J.S., Z.H., and H.H. supervised the project and wrote the manuscript. All authors contributed to discussions and comments on the manuscript.

Notes

The authors declare no competing financial interest.

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